

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION with OpenCV
COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

Code of Conduct:

I _____ Chalamala Sai Harshitha (192424346)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Sai Harshitha

1. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.**
- (b) Identify key features enabling separation.**
- (c) Apply a suitable segmentation technique.**
- (d) Justify your decision logically.**
- (e) Present your response clearly.**

(a) Understanding of the Scenario

The given image contains objects whose pixel intensity values are significantly different from the background. Because of this clear intensity difference, the objects can be separated accurately using an image segmentation technique. The objective is to partition the image into meaningful regions so that the object is isolated from the background for further processing such as object recognition, measurement, or analysis. This is a simple segmentation problem because the distinction between object and background is based only on intensity values.

(b) Identification of Key Issues

The important issues involved in this scenario are:

- The object and background must have distinct grayscale intensity values.
- A suitable threshold value is required for correct separation.
- Improper threshold selection may classify background pixels as object pixels or vice versa.
- Image noise can reduce segmentation accuracy.
- Uniform lighting conditions help achieve better segmentation results.
- The output should clearly separate the foreground object from the background.

(c) Application of Concepts

The most suitable technique for this scenario is **Global Thresholding**, as the object and background can be distinguished based on their intensity values.

Procedure

Step 1: Acquire the grayscale image.

Step 2: Analyze the image histogram to observe the intensity distribution of the object and background.

Step 3: Select an appropriate threshold value (**T**) that separates the two intensity groups.

Step 4: Apply thresholding to every pixel using the following rule:

- If **Pixel Intensity** > **T**, classify the pixel as **Foreground (Object)**.
- If **Pixel Intensity** ≤ **T**, classify the pixel as **Background**.

Step 5: Generate the binary image where:

- White pixels represent the object.
- Black pixels represent the background.

Step 6: Verify that the object is completely separated from the background and that the object boundaries are preserved.

The output is a binary segmented image that clearly highlights the required object while removing the unwanted background.

(d) Decision Making and Justification

Global Thresholding is the most appropriate method because:

- The scenario clearly states that the object is distinguishable from the background based on intensity.
- It is one of the simplest and fastest segmentation techniques.
- It requires less computational time and memory.
- It produces accurate segmentation when the object and background have distinct intensity values.
- It is easier to implement than advanced segmentation methods such as Region Growing, Watershed Segmentation, or Clustering.
- The binary output produced by thresholding is suitable for further computer vision tasks such as object recognition, counting, and feature extraction.

Therefore, Global Thresholding is the best choice for this scenario because it provides high accuracy with minimal computational complexity.

(e) Clear and Structured Conclusion

In this scenario, **Global Thresholding** is the best segmentation technique because it effectively separates the object from the background based on intensity differences. The method is simple, accurate, and efficient, producing a clear binary image that is suitable for subsequent image processing tasks such as feature extraction, object detection, and image analysis.

2. Threshold Selection Problem

An image requires segmentation, but selecting a threshold is difficult.

- (a) Interpret the scenario and explain the challenge.**
- (b) Identify key issues in threshold selection.**
- (c) Apply a suitable thresholding method.**
- (d) Justify your approach logically.**
- (e) Present your explanation clearly.**

(a) Understanding of the Scenario

The objective is to separate the foreground objects from the background using image segmentation. However, selecting a suitable threshold is difficult because the pixel intensity values are not clearly distinguishable. This may result in inaccurate segmentation if a fixed threshold is used.

Key observations:

- The image contains overlapping intensity values.
- Foreground and background are not clearly separated.
- Manual threshold selection is difficult.
- An automatic threshold selection technique is required.

(b) Identification of Key Issues

The major issues in this scenario are:

- Overlapping intensity values between object and background.
- Uneven illumination across the image.
- Presence of image noise.
- Difficulty in selecting a single threshold value.
- Risk of over-segmentation or under-segmentation.
- Reduced segmentation accuracy due to incorrect threshold selection.

(c) Application of Concepts

Suitable Technique: Otsu's Thresholding

Steps

1. Convert the image to grayscale.

2. Compute the image histogram.
3. Analyze the intensity distribution.
4. Automatically calculate the optimum threshold using Otsu's algorithm.
5. Classify pixels into:
 - **Foreground** (object)
 - **Background**
6. Generate the binary segmented image.

Advantages of Otsu's Method:

- Automatic threshold selection.
- No manual intervention required.
- Produces better segmentation for images with two intensity classes.
- Simple and computationally efficient.

(d) Decision Making and Justification

Otsu's Thresholding is the most appropriate technique because:

- It automatically determines the optimal threshold.
- It minimizes classification errors.
- It reduces dependence on user-selected threshold values.
- It performs well when the histogram has two distinct peaks.
- It improves segmentation accuracy.
- It is widely used in document processing, medical imaging, and industrial inspection applications.

(e) Clear Conclusion

In this scenario, **Otsu's Thresholding** is the best solution because it automatically selects the optimum threshold based on the image histogram. It overcomes the difficulty of manual threshold selection, improves segmentation accuracy, and provides a clear separation between the foreground and background. The segmented image can then be effectively used for object detection, feature extraction, and further image analysis.

3. Multiple Object Detection

An image contains multiple objects with similar intensity values.

- (a) Interpret the scenario and explain segmentation difficulty.
- (b) Identify key issues in distinguishing objects.
- (c) Apply a suitable segmentation technique.
- (d) Justify your choice logically.
- (e) Present your explanation clearly.

(a) Understanding of the Scenario

The given image contains **multiple objects whose intensity values are similar to each other and to the background**. Because of this similarity, simple intensity-based segmentation techniques cannot accurately separate individual objects. The objective is to segment each object correctly without merging them into a single region.

Key Observations

- Multiple objects are present in the image.
- Objects have similar grayscale intensity values.
- Intensity alone is insufficient for segmentation.

(b) Identification of Key Issues

The major challenges in this scenario are:

Issue	Explanation
Similar intensity values	Objects cannot be separated using simple thresholding.
Poor object distinction	Adjacent objects may appear as a single region.
Incorrect segmentation	Over-segmentation or under-segmentation may occur.
Touching or overlapping objects	Boundaries between objects become difficult to identify.
Feature similarity	Additional features such as shape, texture, or edges may be required for accurate segmentation.

(c) Application of Concepts

The **Watershed Segmentation** method is suitable because it separates connected or similar objects based on their boundaries.

Procedure

1. Read the input image.

2. Apply a noise removal filter (e.g., Gaussian Filter).
3. Convert the image to grayscale. Detect object edges or markers.
4. Apply the Watershed algorithm to identify individual regions.
5. Separate touching objects using watershed boundaries.
6. Label each detected object for further analysis.

Advantages of Watershed Segmentation

- Separates touching or overlapping objects.
- Produces closed object boundaries.
- Provides accurate object localization.

(d) Decision Making and Justification

The Watershed algorithm is the most appropriate choice because:

- It does not depend only on pixel intensity.
- It accurately separates objects that have similar intensity values.
- It identifies individual object boundaries effectively.
- It reduces merging of adjacent objects.
- It improves segmentation accuracy for complex images.

Comparison of Methods

Technique	Suitable	Reason
Global Thresholding	X No	Cannot separate objects with similar intensities.
Edge Detection	Partial	Detects boundaries but does not perform complete segmentation.
Watershed Segmentation	✓ Yes	Accurately separates touching objects with similar intensity values.

(e) Clear Conclusion

For an image containing multiple objects with similar intensity values, **Watershed Segmentation** is the most suitable technique. It effectively separates connected or overlapping objects by identifying their boundaries instead of relying only on intensity differences. This results in accurate object detection and improved segmentation, making it suitable for applications such as medical imaging, industrial inspection, object recognition, and image analysis.

4. Frequency-Based Enhancement

An image requires enhancement using frequency domain methods.

- (a) Interpret the scenario and explain frequency-based processing.**
- (b) Identify key issues in spatial domain methods.**
- (c) Apply an appropriate frequency domain technique.**
- (d) Justify your choice logically.**
- (e) Present your explanation clearly.**

(a) Understanding of the Scenario

The given image requires enhancement using **frequency domain processing** instead of direct spatial domain operations. In the frequency domain, the image is transformed using the **Fourier Transform**, allowing different frequency components to be processed separately. This helps improve image quality by enhancing important features such as edges or by removing unwanted noise.

Key Observations

- Image enhancement is required.
- Frequency domain techniques are preferred over spatial domain methods.
- The image is transformed into the frequency domain before processing.
- Appropriate filters are used to improve image quality.
- The enhanced image is then transformed back to the spatial domain.

(b) Identification of Key Issues

The major issues in spatial domain enhancement are:

Issue	Explanation
Limited control	Spatial filters directly modify pixel values, making selective enhancement difficult.
Noise amplification	Some spatial filters may also enhance unwanted noise.
Edge preservation	Maintaining important edges while reducing noise is challenging.
Processing efficiency	Complex enhancement operations may require multiple filtering steps.
Frequency separation	Spatial methods cannot directly separate low- and high-frequency components.

(c) Application of Concepts

Suitable Technique: Frequency Domain Filtering using Fourier Transform (FFT)

Procedure

1. Read the input image.
2. Apply the **Fast Fourier Transform (FFT)** to convert the image into the frequency domain.
3. Analyze the frequency spectrum.
4. Select an appropriate filter:
 - **Low-Pass Filter (LPF)** – removes high-frequency noise and smooths the image.
 - **High-Pass Filter (HPF)** – enhances edges and fine details.
5. Apply the selected frequency filter.
6. Perform the **Inverse Fourier Transform (IFFT)** to convert the processed image back to the spatial domain.
7. Obtain the enhanced output image.

(d) Decision Making and Justification

Frequency domain filtering is the most appropriate method because:

- It processes the image based on frequency components rather than individual pixels.
- It provides better flexibility for removing noise and enhancing details.
- Different filters can be selected depending on the enhancement objective.
- It is highly effective for images affected by periodic noise.
- It is widely used in medical imaging, satellite imaging, remote sensing, and industrial inspection.

(e) Clear Conclusion

Frequency domain enhancement is an effective technique for improving image quality by processing the image in the frequency domain. By applying suitable filters such as **Low-Pass Filters** for noise reduction or **High-Pass Filters** for edge enhancement, the image becomes clearer and more informative. This approach provides better control over image enhancement than many spatial domain methods and is widely used in applications such as medical imaging, satellite image processing, industrial inspection, and computer vision.

5. Edge Preservation Filtering

A filtering method is required to remove noise but preserve edges.

- (a) Interpret the scenario and explain the requirement.
- (b) Identify key issues in filtering.
- (c) Apply a suitable technique.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

(a) Understanding of the Scenario

The given image contains unwanted noise that reduces image quality, but the important object edges must be preserved during filtering. Ordinary smoothing filters can remove noise but often blur edges, resulting in the loss of important image details. Therefore, a filtering technique is required that reduces noise while maintaining sharp and clear object boundaries.

Key Observations

- The image is affected by noise.
- Important edges should remain unchanged after filtering.
- Excessive smoothing should be avoided.
- The selected filter must balance noise removal and edge preservation.
- The enhanced image should be suitable for further processing such as segmentation and feature extraction.

(b) Identification of Key Issues

The major challenges in this scenario are:

Issue	Explanation
Noise in the image	Reduces image quality and hides important information.
Edge blurring	Ordinary smoothing filters may blur object boundaries.
Loss of fine details	Important image features may disappear after filtering.
Filter selection	Choosing a filter that removes noise while preserving edges is essential.
Segmentation accuracy	Poor filtering may reduce the accuracy of later image processing tasks.

(c) Application of Concepts

Suitable Technique: Bilateral Filter

The **Bilateral Filter** is the most suitable technique because it smooths the image while preserving sharp edges.

Procedure

1. Read the input image.
2. Identify the noisy regions in the image.
3. Apply the **Bilateral Filter**.
4. The filter considers:
 - **Spatial distance** between neighboring pixels.
 - **Intensity difference** between neighboring pixels.
5. Smooth pixels having similar intensity values.
6. Preserve pixels across strong edges.
7. Obtain the enhanced image with reduced noise and clear boundaries.

(d) Decision Making and Justification

The **Bilateral Filter** is the most appropriate choice because:

- It removes noise while maintaining edge sharpness.
- It considers both pixel distance and intensity similarity during filtering.
- It avoids excessive smoothing compared to averaging or Gaussian filters.
- It improves image quality without losing important structural information.
- It is widely used in computer vision, medical imaging, photography, and image enhancement applications.

(e) Clear Conclusion

For an image where noise must be removed without affecting important edges, the **Bilateral Filter** is the most suitable filtering technique. It effectively smooths homogeneous regions while preserving edge information by considering both spatial proximity and intensity similarity. As a result, the enhanced image has reduced noise, sharper boundaries, and better visual quality, making it highly suitable for segmentation, feature extraction, object detection, and other computer vision applications.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand